

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech.

Branch: Common to All branches

Semester: I

Subject Code & Name:

Engineering Physics (BTBS102P)

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The peak of resonance _____ with increase in damping. a. increases b. decreases c. remains constant d. no relation	CO 1	1
2	In the absence of damping, the motion of a harmonic oscillator is a. Overdamped b. Underdamped c. Critically damped d. Simple harmonic motion	CO 1	1
3	In a thin-film interference, colors are observed due to a. Absorption of light b. reflection and refraction of light c. scattering of light d. diffraction of light	CO 2	1
4	Formula for numerical aperture is a. $\sqrt{n_1^2 - n_2^2}$ b. $\sqrt{n_1^2 + n_2^2}$ c. $\sqrt{n_1 - n_2}$ d. $n_1 + n_2$	CO 2	1
5	The Hall Voltage developed in Hall effect depends on a. Magnetic field b. Current Density c. Magnetic Field and Current density d. Material resistance	CO 5	1
6	$\psi \psi^*$ is a. Probability b. Wave function c. Wave Particle duality d. Electron mass	CO 3	1

7	The temperature below which the material becomes superconductor is called _____				CO 5	1
	a. Curie Temperature	b. Neel Temperature	c. Critical Temperature	d. Conducting temperature		
8	When the population of lower energy level is LESS than the population at higher energy level, it is called,				CO 2	1
	a. Population density	b. Population inversion	c. population diversion	d. population depletion		
9	Miller indices represents				CO 4	1
	a. One unit cell	b. One given plane	c. set of parallel planes	d. Lattice structure		
10	The angle of refraction in total internal reflection is,				CO 2	1
	a. $<90^\circ$	b. $>90^\circ$	c. $=90^\circ$	d. all of these		
11	The statement for Heisenberg's uncertainty Principle is				CO 3	1
	a. $\Delta x. \Delta p \geq \frac{h}{2\pi}$	b. $\Delta x. \Delta p \gg \frac{h}{2\pi}$	c. $\Delta x. \Delta p \leq \frac{h}{2\pi}$	d. $\Delta x. \Delta p \ll \frac{h}{2\pi}$		
12	Coordination number 12 is of _____ Cubic Crystal structure.				CO 4	1
	a. SC	b. BCC	c. FCC	d. All of these		
Q.2	Solve the following.					12
A)	Describe the construction and working of piezoelectric oscillator to produce ultrasonic waves. Calculate the natural frequency of 40 mm pure iron rod. Given the density of pure iron is $7.25 \times 10^3 \text{ kg/m}^3$ and Young's modulus $115 \times 10^9 \text{ N/m}^2$.				CO 1	6
B)	What are free oscillations? Derive the mathematical expression for free oscillation.				CO 1	6
Q.3	Solve the following.					12
A)	Describe the construction and working of He-Ne Laser with neat and labeled diagram.				CO 2	6
B)	Derive an expression for darkness of a thin film. Calculate the minimum thickness of a thin film which appears dark, when viewed under the light of wavelength 600 nm and refractive index of thin				CO 2	6

	film is 1.5		
Q. 4	Solve Any Two of the following.		12
A)	Derive Schrodinger's Time independent Equation.	CO 3	6
B)	Prove on the basis of Heisenberg's uncertainty Principle, electron cannot reside inside the nucleus.	CO 3	6
C)	What is wave function? Explain its significance. Derive an expression for Schrodinger's Time dependent Equation.	CO 3	6
Q.5	Solve Any Two of the following.		12
A)	Derive an expression for atomic packing fraction for BCC and FCC crystal structure.	CO 4	6
B)	Explain with neat and labelled diagram the production of Line and continuous X-Rays.	CO 4	6
C)	Describe construction and working of GM Counter.	CO 4	6
Q. 6	Solve Any Two of the following.		12
A)	Distinguish between Type I and Type II Superconductors along with examples.	CO 5	6
B)	Derive an expression for Hall coefficient, with neat and labeled diagram.	CO 5	6
C)	Define magnetic domain. Discuss various types of magnetic materials with domain structures.	CO 5	6
	*** End ***		